Genome-Informed Metagenomic Recruitment and Comparative Genomics Identify a Recurrently Detected Eelgrass-Associated *Vibrio* Lineage Across Northern Hemisphere Sites

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Background

• Eelgrass-associated bacteria are often analyzed at the genus level, obscuring lineage information

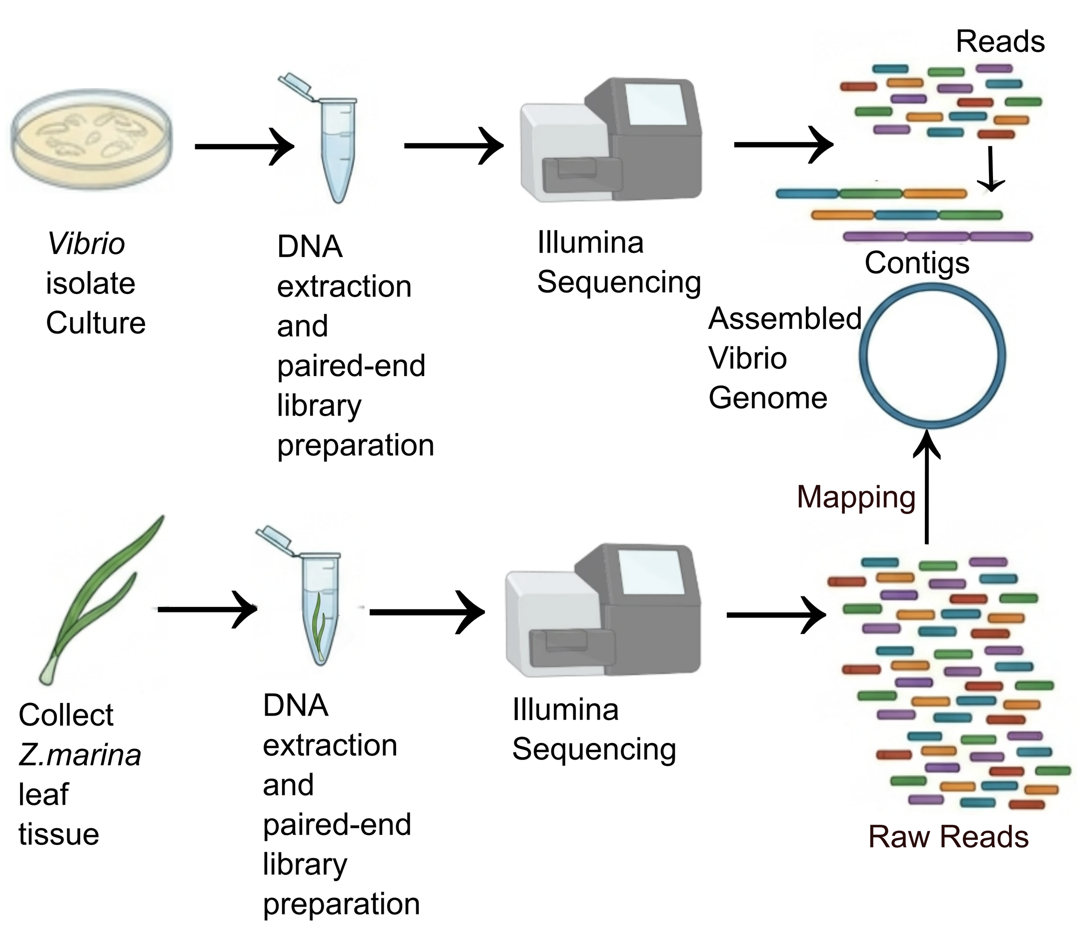
• Genome-informed metagenomic recruitment approaches can improve lineage studies by linking isolate genomes to environmental metagenomic reads.

• Our goal was to identify which eelgrass-associated *Vibrio* isolates cultured from Bodega Bay were repeatedly recovered across sites from our global metagenomic dataset

Methods

• Nineteen eelgrass-associated *Vibrio* isolates were cultured and sequenced to generate draft genomes. Genomic DNA was extracted from each isolate, prepared for Illumina short-read sequencing, and assembled after standard quality control.

• For environm ntal recruitment analysis, *Zostera marina* leaf-associated microbial communities were sampled and subjected to Illumina metagenomic sequencing. Quality-filtered metagenomic reads were mapped against the isolate genome set using Bowtie2, and recruitment was quantified as relative abundance and genome coverage across sites.

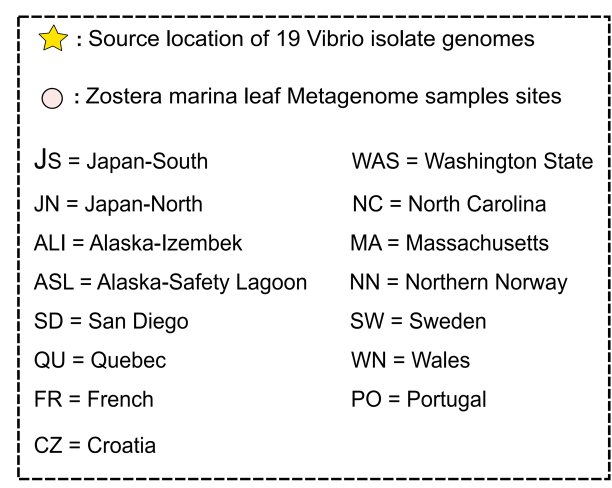


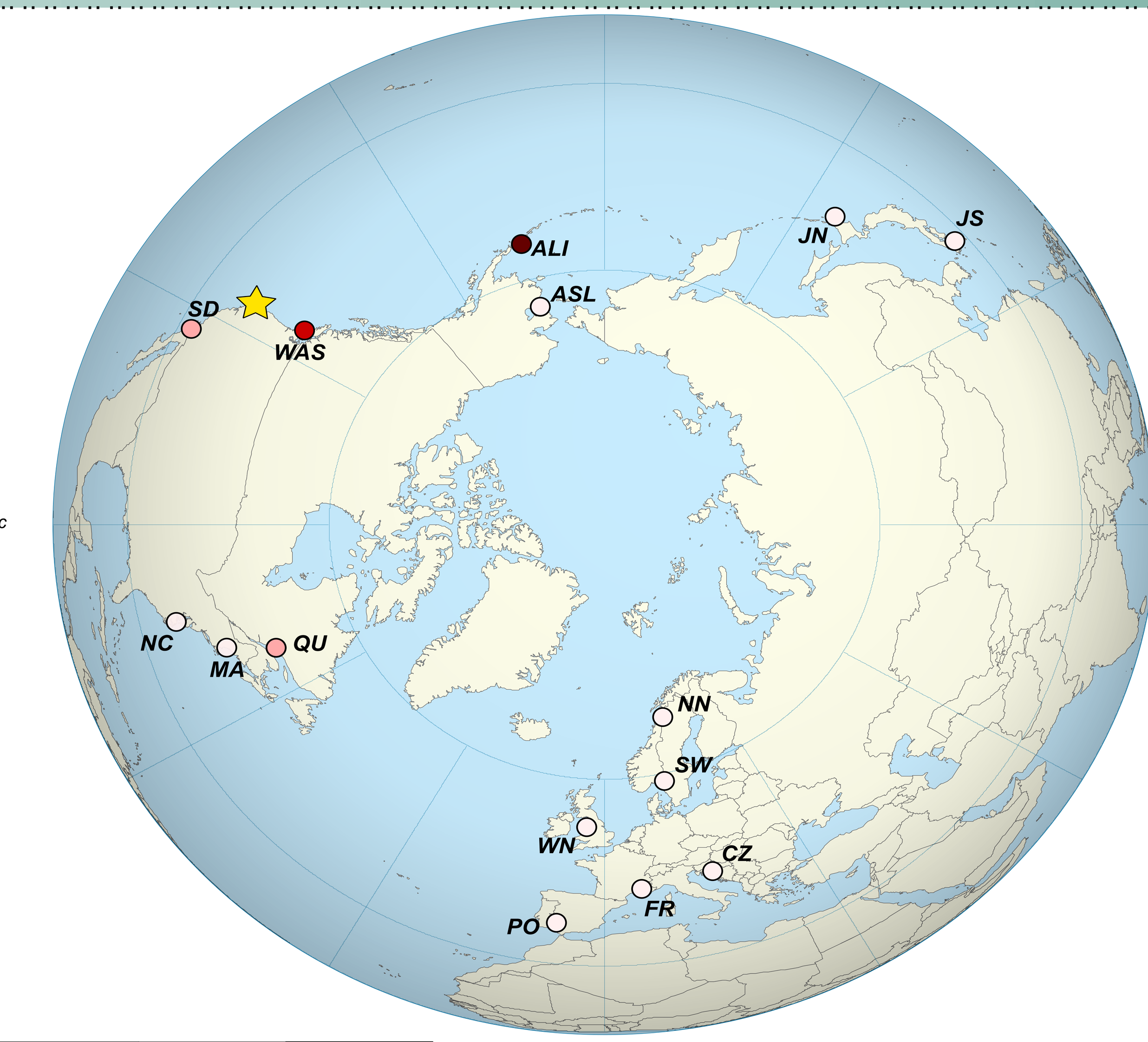
• Genome quality was evaluated with CheckM2, and taxonomic placement was assessed using GTDB ANI screening. Gene content was functionally annotated, and selected isolate was compared with closely related Vibrio genomes in a pangenome framework to identify candidate gene-content differences associated with its ecological distribution.

Result

Geographic Site	Mapped Reads(%)	Recruitment Status
ALI*	8.723%	Strong recruitment
WAS	0.289%	Moderate recruitment
QU	0.008%	Low recruitment
SD	0.005%	Low recruitment
Other Sites(n=11)*	<0.001%	Minimal or undetectable recruitment

**MA,NC,JN,JS,ASL,NN,SW,FR,PO,CZ and WN were all below 0.001% mapped read*
**ALI's unusually high recruitment may be partly influenced by damaged host tissue, although a site-specific geographic effect cannot be ruled out.*





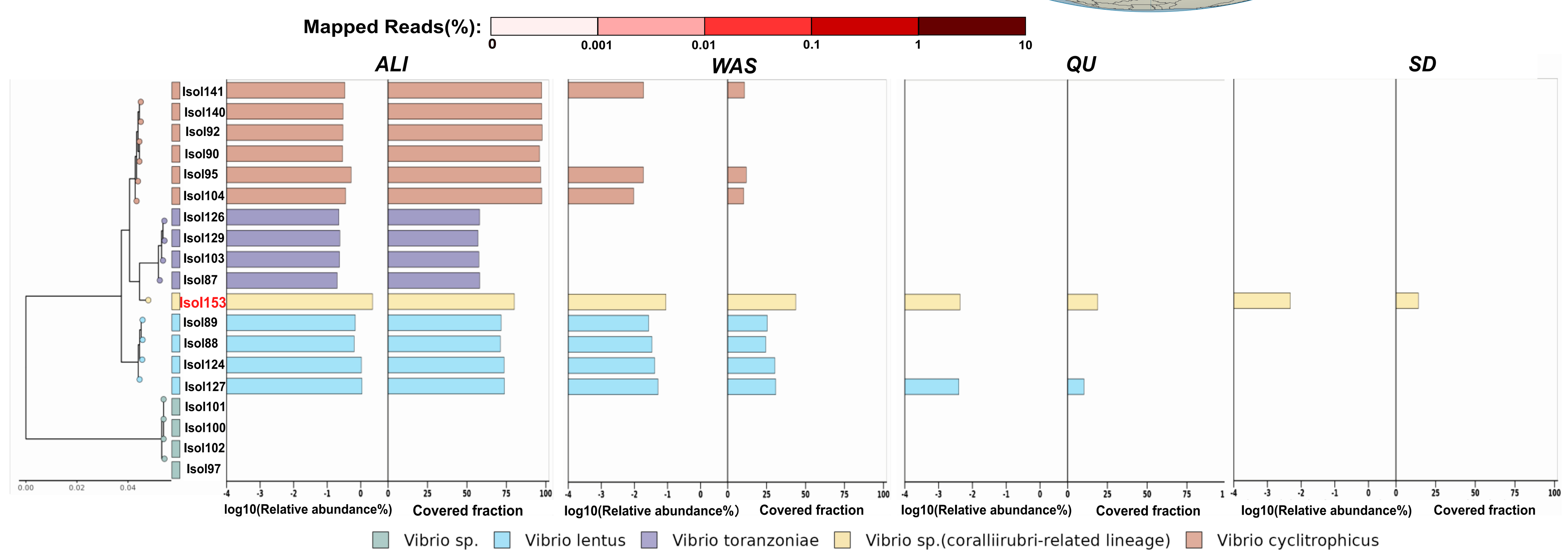
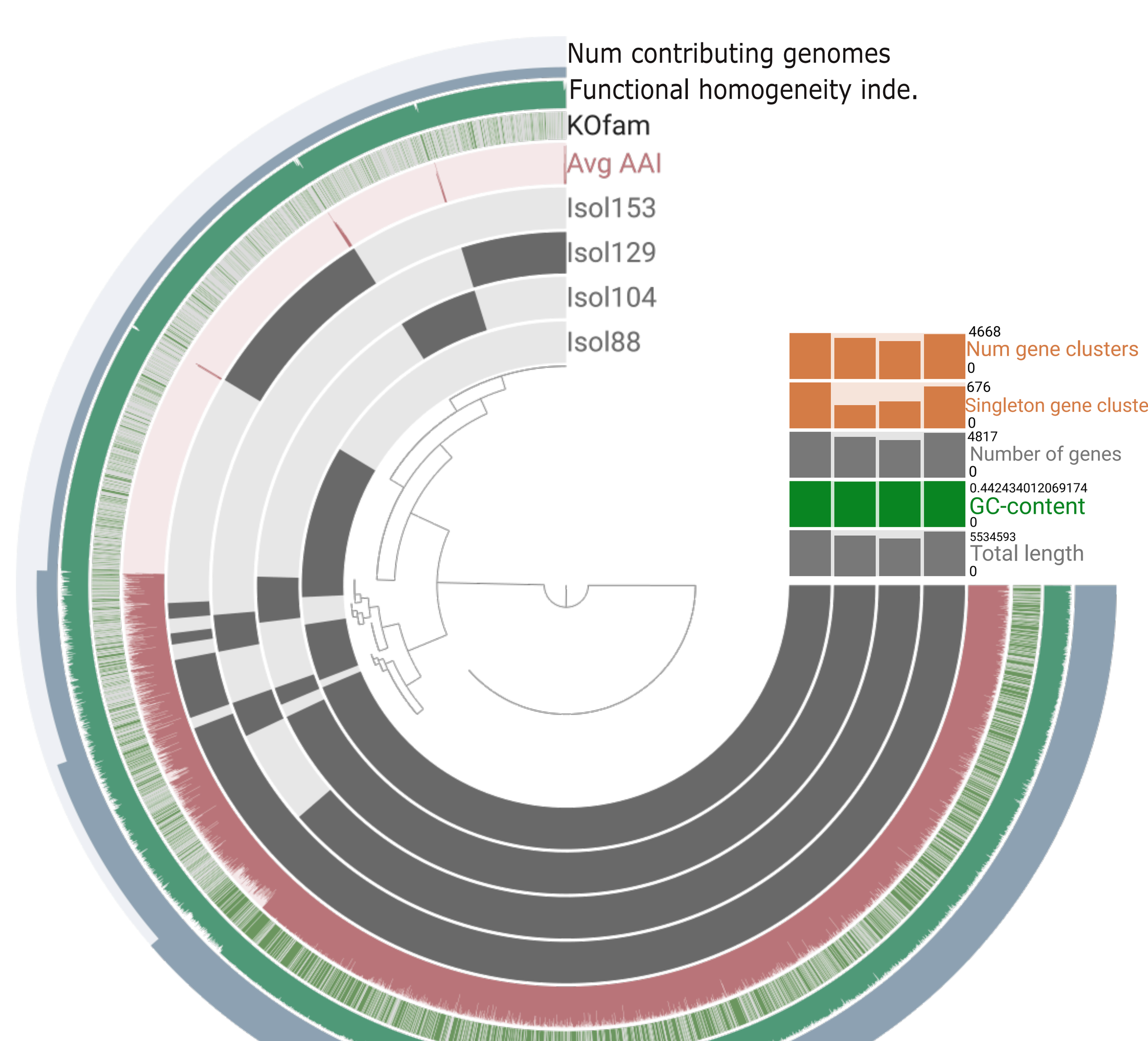


Figure 1. Geographic distribution and metagenomic recruitment of eelgrass-associated *Vibrio* isolates across Northern Hemisphere sites The map summarizes the geographic distribution of metagenomic sampling and the recruitment panel shows the relative abundance and genome coverage of cultured representative isolates. Isolate 153 (in red) showed the strongest and most consistent recruitment across sites.



Isol153

Genus	*Vibrio*
Closest GTDB species	*Vibrio coralliirubri*
ANI to closest reference	**94.76%**
Alignment fraction (AF)	0.771
GTDB species circumscription satisfied	No
Interpretation	Not confidently assigned to an existing GTDB species cluster

Figure 2. Pangenome comparison of Isol153 and closely related *Vibrio* isolates. The pangenome structure shows that Isol153 shares a conserved genomic backbone with nearby isolates while also carrying substantial variable gene-cluster content concentrated in accessory and singleton-like regions. These lineage-specific differences suggest that Isol153 is closely related to the compared isolates at the core-genome level, but may differ in flexible genomic regions that contribute to ecological or functional variation.

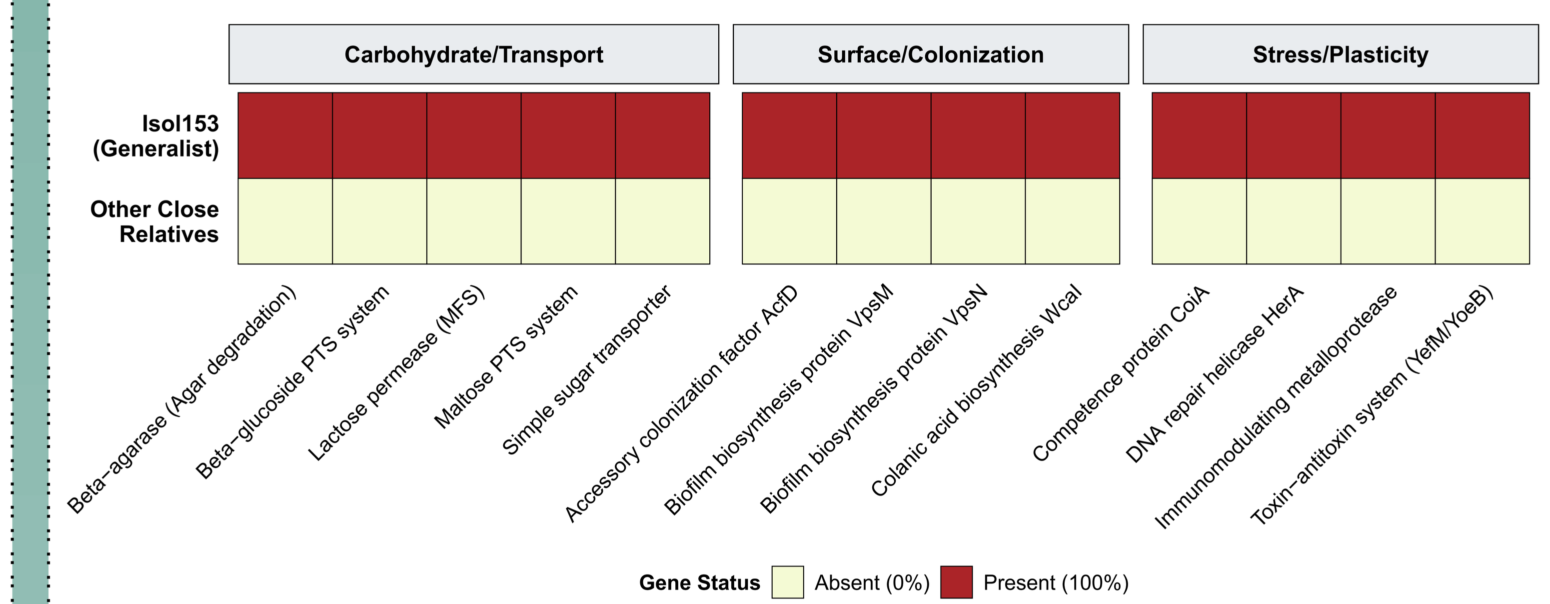


Figure 3. Isol153-specific candidate functional modules relative to closely related isolates. Comparative KOfam-based functional profiling showed that Isol153 carries a distinct set of candidate functions absent from closely related isolates. These functions group into three major categories—carbohydrate/transport, surface colonization, and stress/plasticity—including beta-agarase, sugar transport systems, colonization and biofilm-associated proteins, and DNA repair/stress-related features, suggesting that Isol153 may combine broader substrate use with enhanced surface-associated persistence.

Conclusion

• Genome-informed metagenomic recruitment resolved eelgrass-associated *Vibrio* diversity at the lineage level across geographically distant *Zostera marina* sites.

• Isol153 showed the strongest and most consistent recruitment across focal metagenomic samples, suggesting that it may represent a broadly distributed eelgrass-associated lineage.

• ANI comparison showed that Isol153 is genomically distinct from currently defined GTDB species clusters. Pangenome and functional analyses suggest that Isol153 carries accessory gene content related to carbohydrate utilization, surface colonization, and stress response.

• Together, these results identify Isol153 as a strong candidate for future studies of eelgrass-associated microbial adaptation and biogeographic distribution.

Reference

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